Bernoulli Naive Bayes:

This is similar to the multinomial naive bayes but the predictors are boolean variables. The parameters that we use topredict the class variable take up only values yes or no, for example if a word occurs in the text or not.

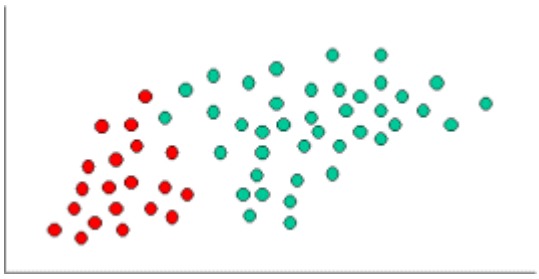
Gaussian Naive Bayes:

When the predictors take up a continuous value and are not discrete, we assume that these values are sampled from agaussian distribution.

Conclusion:

Naive Bayes algorithms are mostly used in sentiment analysis, spam filtering, recommendation systems etc. They arefast and easy to implement but their biggest disadvantage is that the requirement of predictors to be independent. In most of the real life cases, the predictors are dependent, this hinders the performance of the classifier.

Naive Bayes Classifier technique is based on the so-called Bayesian theorem and is particularly suited when the dimensionality of the inputs is high. Despite its simplicity, Naive Bayes can often outperform more sophisticated classification methods.



To demonstrate the concept of Naïve Bayes Classification, consider the example displayed in the illustration above. As indicated, the objects can be classified as either GREEN or RED. Our task is to classify new cases as they arrive, i.e., decide to which class label they belong, based on the currently exiting objects.

Since there are twice as many GREEN objects as RED, it is reasonable to believe that a new case (which hasn't been observed yet) is twice as likely to have membership GREEN rather than RED. In the Bayesian analysis, this belief isknown as the prior probability. Prior probabilities are based on previous experience, in this case the percentage of GREEN and RED objects, and often used to predict outcomes before they actually happen.
Thus, we can write:

Although the prior probabilities indicate that X may belong to GREEN (given that there are twice as many GREEN compared to RED) the likelihood indicates otherwise; that the class membership of X is RED (given that there are more RED objects in the vicinity of X than GREEN). In the Bayesian analysis, the final classification is produced by combining both sources of information, i.e., the prior and the likelihood, to form a posterior probability using the so-called Bayes' rule (named after Rev. Thomas Bayes 1702-1761).

$$Posterior\ probability\ of\ X\ being\ GREEN \propto$$
$$Prior\ probability\ of\ GREEN \times Likelihood\ of\ X\ given\ GREEN$$
$$= \frac{4}{6} \times \frac{1}{40} = \frac{1}{60}$$
$$Posterior\ probability\ of\ X\ being\ RED \propto$$
$$Prior\ probability\ of\ RED \times Likelihood\ of\ X\ given\ RED$$
$$= \frac{2}{6} \times \frac{3}{20} = \frac{1}{20}$$

Assume that you are given a characteristic information of 10,000 people living in your town. You are asked to study them and come up with the algorithm which should be able to tell whether a new person coming to the town is male or a female.

Primarily you are given information about:

Skin colourHair lengthWeight Height
Based on the information you can divide the information in such a way that it somehow indicates the characteristicsof Males vs. Females.

Below is a hypothetical tree designed out of this data:

The tree shown above divides the data in such a way that we gain the maximum information, to understand the tree
– If a person's hair length is less than 5 Inches, weight greater than 55 KGs then there are 80% chances for thatperson being a Male.

If you are familiar with Predictive Modelling e.g., Logistic Regression, Random Forest etc. – You might be wondering what is the difference between a Logistic Model and Decision Tree! Because in both the algorithms we are trying to predict a categorical variable.

There are a few fundamental differences between both but ideally both the approaches should give you the same results. The best use of Decision Trees is when your solution requires a representation. For example, you are working for a Telecom Operator and building a solution using which a call center agent can take a

decision whether to pitch for an upsell or not!

There are very less chances that a call center executive will understand the Logistic Regression or the equations, but using a more visually appealing solution you might gain a better adoption from your call center team.

How does Decision Tree work?

There are multiple algorithms written to build a decision tree, which can be used according to the problem characteristics you are trying to solve. Few of the commonly used algorithms are listed below:

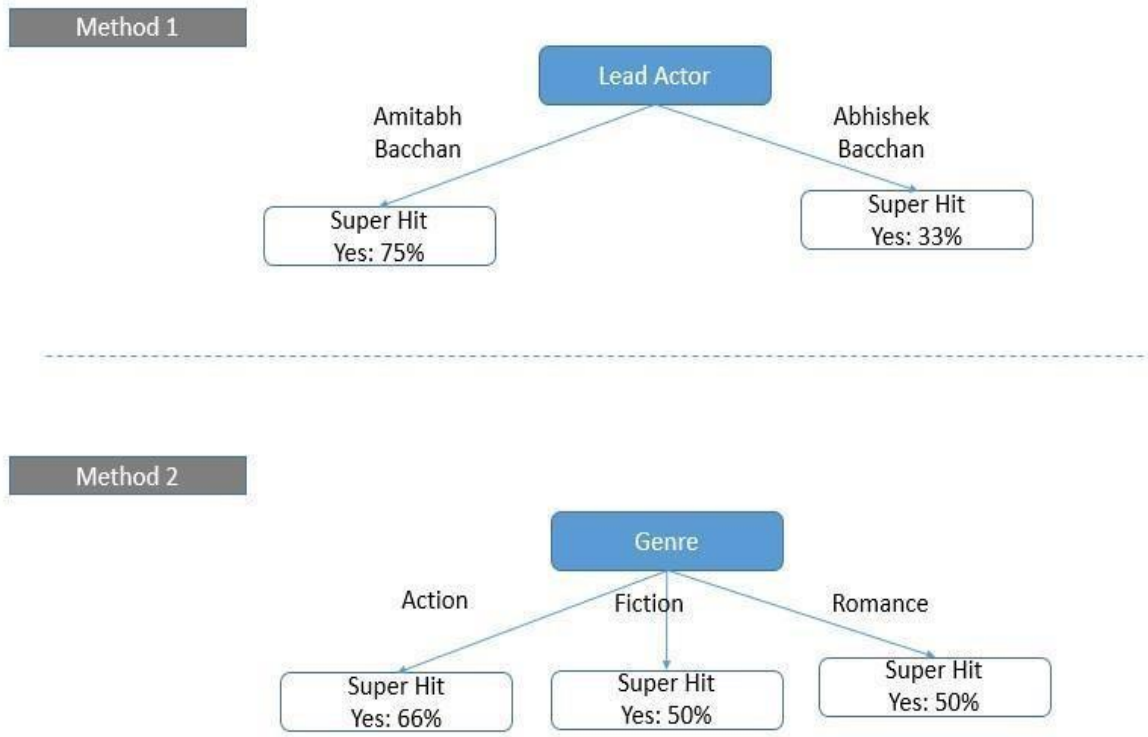
ID3 C4.5 CART
CHAID (CHi-squared Automatic Interaction Detector)MARS
Conditional Inference Trees

Though the methods are different for different decision tree building algorithms but all of them works on the principle of Greediness. Algorithms try to search for a variable which give the maximum information gain or divides the data in the most homogenous way.

For an example, consider the following hypothetical dataset which contains Lead Actor and Genre of a movie alongwith the success on box office:

Lead Actor	**Genre**	**Hit(Y/N)**
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
Abhishek Bacchan	*Action*	*No*
Abhishek Bacchan	*Fiction*	*No*
Abhishek Bacchan	*Romance*	*Yes*

Let say, you want to identify the success of the movie but you can use only one variable – There are the followingtwo ways in which this can be done:



You can clearly observe that Method 1 (Based on lead actor) splits the data best while the second method (Based on Genre) have produced mixed results. Decision Tree algorithms do similar things when it comes to select variables.

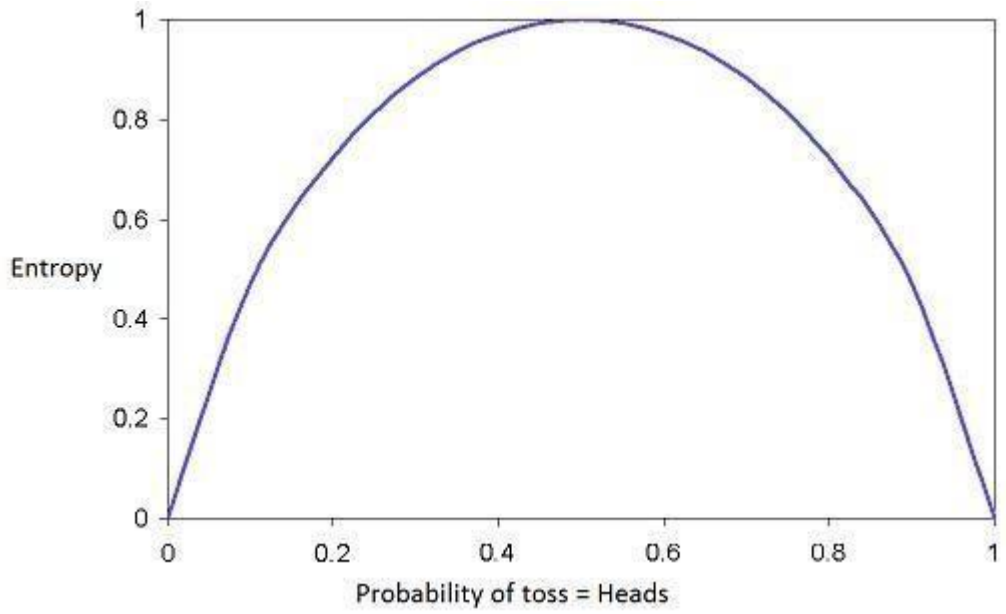
There are various metrics which decision trees use in order to find out the best split variables. We'll go through them one by one and try to understand, what do they mean?

Entropy & Information Gain

The word Entropy is borrowed from Thermodynamics which is a measure of variability or chaos or randomness. Shannon extended the thermodynamic entropy concept in 1948 and introduced it into statistical studies and suggested the following formula for statistical entropy:

$$H = -\sum_{j=1}^{n} P_j ln P_j$$

Where, H is the entropy in the system which is a measure of randomness.
Assuming you are rolling a fair coin and want to know the Entropy of the system. As per the formula given by Shann – Entropy would be equals to *-[0.5 ln(0.5) + 0.5 ln(0.5)]*.
Which is equal to -0.69; which is the maximum entropy which can occur in the system. In other words, there will be maximum randomness in our dataset if the probable outcomes have same probability of occurrence.



Graph shown above shows the variation of Entropy with the probability of a class, we can clearly see that Entropy ismaximum when probability of either of the classes is equal. Now, you can understand that when a decision algorithm tries to split the data, it selects the variable which will give us maximum reduction in system Entropy.

For the example of movie success rate – Initial Entropy in the system was:

*EntropyParent = -(0.57*ln(0.57) + 0.43*ln(0.43));* Which is 0.68

Entropy after Method 1 Split

Entropyleft = -(.75*ln(0.75) + 0.25*ln(0.25)) =

Entropyright = -(.33*ln(0.33) + 0.67*ln(0.67)) = 0.63

Captured impurity or entropy after splitting data using Method 1 can be calculated using the followingformula: "*Entropy (Parent) – Weighted Average of Children Entropy*"
Which is,

*0.68 – (4*0.56 + 3*0.63)/7 = 0.09*
This number 0.09 is generally known as "Information Gain"

Entropy after Method 2 Split

*Entropyleft = -(.67*ln(0.67) + 0.33*ln(0.33)) =*

*Entropymiddle = -(.5*ln(0.5) + 0.5*ln(0.5)) =*

*Entropyright = -(.5*ln(0.5) + 0.5*ln(0.5)) = 0.69*

Now using the method used above, we can calculate the Information Gain as:

*Information Gain = 0.68 – (3*0.63 + 2*0.69 + 2*0.69)/7 = 0.02*

Hence, we can clearly see that Method 1 gives us more than 4 times information gain compared to Method 2 and hence Method 1 is the best split variable.

Gain Ratio

Soon after the development of entropy mathematicians realized that Information gain is biased toward multi-valued attributes and to conquer this issue, "Gain Ratio" came into picture which is more reliable than Information gain. The gain ratio can be defined as:

$$Gini_S(K) = \frac{T_1}{T}\, Gini\,(T_1) + \frac{T_2}{T}\, Gini\,(T_2)$$

Where Split info can be defined as:

$$-\sum_{i=1}^{n} Di\ log_2 Di$$

Assuming we are dividing our variable into 'n' child nodes and Di represents the number of records going into various child nodes. Hence gain ratio takes care of distribution bias while building a decision tree.

For the example discussed above, for Method 1

*Split Info = - ((4/7)*log2(4/7)) - ((3/7)*log2(3/7)) = 0.98*

And Hence,

Gain Ratio = 0.09/0.98 = 0.092

Gini Index

There is one more metric which can be used while building a decision tree is Gini Index (Gini Index is mostly used in CART). Gini index measures the impurity of a data partition K, formula for Gini Index can be written down as:

$$Gini(K) = 1 - \sum_{i=1}^{n} {P_i}^2$$

Where m is the number of classes, and P_i is the probability that an observation in K belongs to the class. Gini Index assumes a binary split for each of the attribute in S, let say T_1 & T_2. The Gini index of K given this partitioning is given by:

Which is nothing but a weighted sum of each of the impurities in split nodes. The reduction in impurity is given by:

$$Gini(K) - Gini_S(K)$$

Similar to Information Gain & Gain Ratio, split which gives us maximum reduction in impurity is considered fordividing our data.

Coming back to our movie example,If we want to calculate *Gini(K)*-

= 0.49

Now as per our Method 1, we can get *Gini$_S$(K)* as,

$$Gini_S(K) = \frac{T_1}{T}\, Gini\,(T_1) + \frac{T_2}{T}\, Gini\,(T_2)$$

$$= \frac{4}{7}\left\{1 - \left(\left[\frac{3}{4}\right]^2 + \left[\frac{1}{4}\right]^2\right)\right\} + \frac{3}{7}\left\{1 - \left(\left[\frac{1}{3}\right]^2 + \left[\frac{2}{3}\right]^2\right)\right\}$$

= 0.24 + 0.19

= 0.43

LINEAR REGRESSION

Linear regression is a statistical approach for modelling relationship between a dependent variable with a given setof independent variables.

$$Gini\,(K) = 1 - \left(\left[\frac{4}{7}\right]^2 + \left[\frac{3}{7}\right]^2\right)$$

an approach for predicting a **response** using a **single feature**.

It is assumed that the two variables are linearly related. Hence, we try to find a linear function that predicts theresponse value(y) as accurately as possible as a function of the feature or independent variable(x).

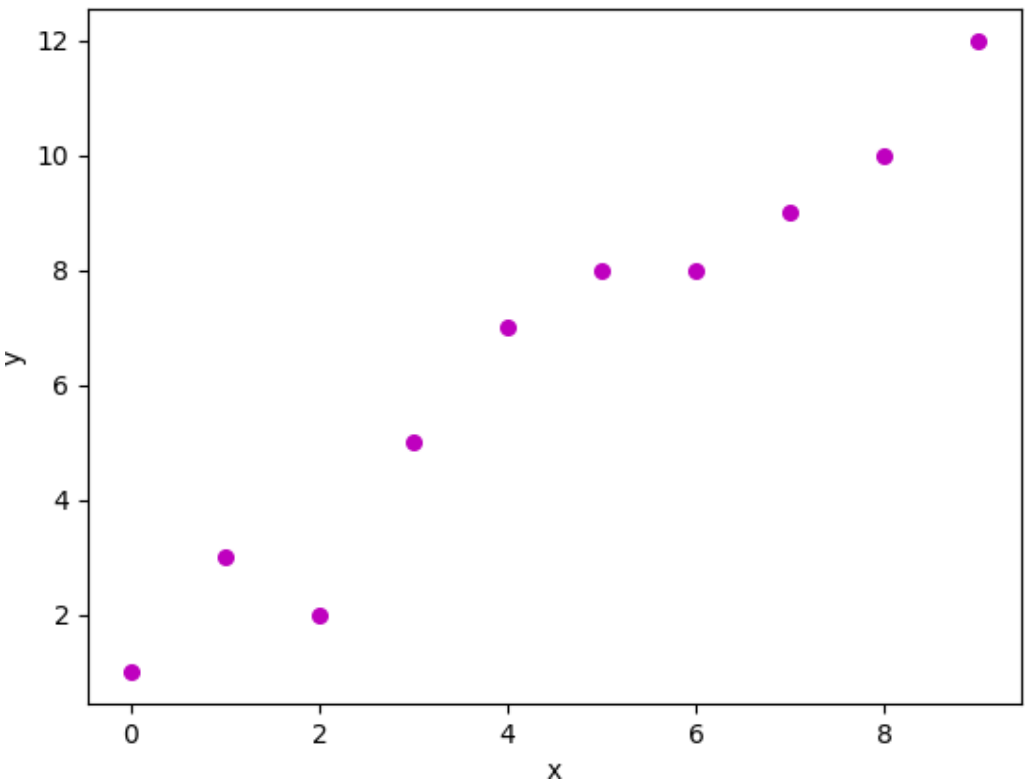
Let us consider a dataset where we have a value of response y for every feature x:

x	0	1	2	3	4	5	6	7	8	9
y	1	3	2	5	7	8	8	9	10	12

For generality, we define:

x as **feature vector**, i.e x = [x_1, x_2, …., x_n], y as **response vector**, i.e y = [y_1, y_2, …., y_n]for **n** observations (in above example, n=10).

A scatter plot of above dataset looks like:-



Now, the task is to find a **line which fits best** in above scatter plot so that we can predict the response for any newfeature values. (i.e a value of x not present in dataset)
This line is called **regression line**.
The equation of regression line is represented as:

Here,

- h(x_i) represents the **predicted response value** for ith observation.
- b_0 and b_1 are regression coefficients and represent **y-intercept** and **slope** of regression line respectively.

To create our model, we must "learn" or estimate the values of regression coefficients b_0 and b_1.
And once we'veestimated these coefficients, we can use the model to predict responses!